

Geometry
Unit 8
Lesson 6 Practice

Name _____
Date _____
Period _____

Two congruent angles of an isosceles trapezoid have measures $(8x + 14)^\circ$ and $(12x - 8)^\circ$.

1. What is the value of x ?

2. What are the measures of each angle in the trapezoid?

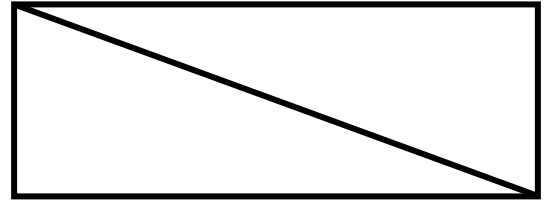
For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of congruent sides and consecutive angles are supplementary.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

4. Quadrilateral $BNTV$ has four congruent sides and diagonals are perpendicular to each other.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

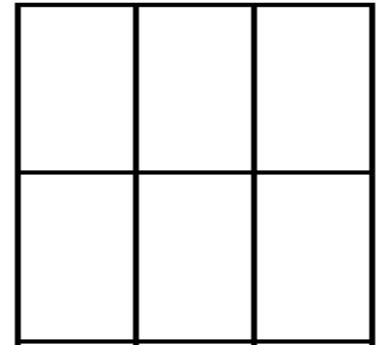
5. Quadrilateral $KPSX$ has two pairs of congruent sides and one pair of opposite congruent angles.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $32x$ feet, width of $46x$ feet and the diagonal of the building is 200 feet.



6. What is the value of x ? Round answers to the nearest hundredth.
7. What is the height of the building? Round answers to the nearest hundredth.
8. What is the width of the building? Round answers to the nearest hundredth.

A wall paneling with a total area of 18 square feet is made up of 6 rectangles, as shown. There are two congruent rectangles along the height and three across the width. Each rectangle in the paneling is $(3x + 1)$ feet in height and $(2x - 1)$ feet in width.



9. What is the value of x ? Round your answer to the nearest hundredth.
10. What are the dimensions of the portion of the wall covered by the paneling?